

Project Report: PDF Q&A Chatbot with RAG - Generative AI Engineer

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Family:	Generative AI Engineer
Level:	Beginner
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Upload PDFs, chunk and embed with OpenAI/HuggingFace, answer questions with retrieved context.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

PDF Q&A Chatbot with RAG** is a beginner-level end-to-end project for Generative AI Engineer. **Duration:** 2-3 weeks. Upload PDFs, chunk and embed with OpenAI/HuggingFace, answer questions with retrieved context. **Tools:** LangChain, OpenAI, Chroma, Streamlit.

Code Reference:

```
Objective: PDF text extraction and chunking
Objective: Vector store (Chroma/FAISS)
Objective: RAG prompt pipeline
Objective: Simple Streamlit or Gradio UI
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Generative AI Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Generative AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Ingest [Beginner]

Chapter focus: Phase: Ingest** *(Level: Beginner)

PDF ? chunks. Complete these tasks: LangChain loader; Splitter config. Deliverable for this phase: Indexed documents.

Code Reference:

- [] LangChain loader
- [] Splitter config

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Ingest is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Indexed documents to the teacher.

Chapter 3: Phase: Embed [Beginner]

Chapter focus: Phase: Embed** *(Level: Beginner)

Vector store setup. Complete these tasks: OpenAI embeddings; Similarity search. Deliverable for this phase: Working retrieval.

Code Reference:

- [] OpenAI embeddings
- [] Similarity search

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Embed is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Working retrieval to the teacher.

Chapter 4: Phase: RAG [Beginner]

Chapter focus: Phase: RAG *(Level: Beginner)**

Prompt with context. Complete these tasks: System prompt; Source citations. Deliverable for this phase: Q&A demo.

Code Reference:

- [] System prompt
- [] Source citations

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase RAG is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Q&A demo to the teacher.

Chapter 5: Phase: UI [Beginner]

Chapter focus: Phase: UI *(Level: Beginner)**

Streamlit chat interface. Complete these tasks: Upload PDF; Chat history. Deliverable for this phase: Deployed app URL.

Code Reference:

- [] Upload PDF
- [] Chat history

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase UI is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Generative AI Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Deployed app URL to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review** *(Level: Beginner)

Submit all final deliverables: GitHub repo; Demo video; Sample Q&A log. Skills demonstrated: RAG, Embeddings, LLM APIs. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: GitHub repo
Deliverable: Demo video
Deliverable: Sample Q&A log

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits GitHub repo and passes the rubric.